

Exercice 32:

$$\begin{aligned} z_1 + z_2 &= 3\alpha + 3 + i(\alpha^2 + 1) + \alpha^2 - \alpha + i(\alpha^2 - 1) \\ &= \alpha^2 + 2\alpha - 3 + i2\alpha^2. \end{aligned}$$

$$\begin{aligned} 1) \quad z_1 + z_2 \in i\mathbb{R} &\Leftrightarrow \operatorname{Re}(z_1 + z_2) = 0 \\ &\Leftrightarrow \alpha^2 + 2\alpha - 3 = 0 \end{aligned}$$

Le discriminant de $\alpha^2 + 2\alpha - 3 = 0$ est :

$$\Delta = 2^2 - 4 \times 1 \times (-3) = 16 > 0$$

Donc :

$$\begin{aligned} z_1 + z_2 \in i\mathbb{R} &\Leftrightarrow \alpha = \frac{-2 - \sqrt{16}}{2 \times 1} \quad \text{ou} \quad \alpha = \frac{-2 + \sqrt{16}}{2 \times 1} \\ &\Leftrightarrow \alpha = -3 \quad \text{ou} \quad \alpha = 1 \end{aligned}$$

Donc $z_1 + z_2$ est imaginaire pur si : $\alpha \in \{-3; 1\}$

$$\begin{aligned} 2) \quad z_1 + z_2 \in \mathbb{R} &\Leftrightarrow \operatorname{Im}(z_1 + z_2) = 0 \\ &\Leftrightarrow 2\alpha^2 = 0 \\ &\Leftrightarrow \alpha = 0 \end{aligned}$$

Donc $z_1 + z_2$ est réel si $\alpha = 0$.